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The impact of DNA double-strand break repair pathways throughout the carbon ion spread-out Bragg peak beam

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Abstract

Following carbon ion beam irradiation in mammalian cells, such as used in carbon ion radiotherapy (CIRT), it has been suggested that the balance between whether nonhomologous end joining (NHEJ) or homologous recombination (HR) is utilized depends on the DNA double-strand break (DSB) complexity. Here, we quantified DSB distribution and identified the importance of each DSB repair pathway at increasing depths within the carbon ion spread-out Bragg peak (SOBP) beam range. Chinese hamster ovary (CHO) cell lines were irradiated in a single biological system capable of incorporating the full carbon ion SOBP beam range. Cytotoxicity and DSB distribution/repair kinetics were examined at increasing beam depths using cell survival as an endpoint and γ-H2AX as a surrogate marker for DSBs. We observed that proximal SOBP had the highest number of total foci/cell and lowest survival, while distal SOBP had the most dense tracks. Both NHEJ- and HR-deficient CHO cells portrayed an increase in radiosensitivity throughout the full carbon beam range, although NHEJ-deficient cells were the most radiosensitive cell line from beam entrance up to proximal SOBP and demonstrated a dose-dependent decrease in ability to repair DSBs. In contrast, HR-deficient cells had the greatest ratio of survival fraction at entrance depth to the lowest survival fraction within the SOBP and demonstrated a linear energy transfer (LET)-dependent decrease in ability to repair DSBs. Collectively, our results provide insight into treatment planning and potential targets to inhibit, as HR was a more beneficial pathway to inhibit than NHEJ to enhance the cell killing effect of CIRT in targeted tumor cells within the SOBP while maintaining limited unwanted damage to surrounding healthy cells.

KEYWORDS
Bragg peak, carbon ion radiotherapy, DNA damage and repair, gamma-H2AX, HR, NHEJ

Abbreviations: CHO, Chinese hamster ovary; CIRT, carbon ion radiotherapy; DSB, DNA double-strand break; HR, homologous recombination; LET, linear energy transfer; NHEJ, nonhomologous end joining; SOBP, spread-out Bragg peak.

[Corrections made on 05 October 2023, after first online publication: One of the co-authors' name was corrected from "Liu Cuihua" to "Cuihua Liu".]

1 | INTRODUCTION

DNA double-strand breaks (DSBs) are known to be the major factor responsible for ionizing radiation-induced cell death when left unrepaired or misrepaired.[1] In mammalian cells, the two main DSB repair pathways are homologous recombination (HR) and nonhomologous end joining (NHEJ).[2] However, for high-LET carbon ion irradiation-induced DSBs, the most important DSB repair pathway choice has been an area of debate throughout the radiobiology community. Following carbon ion irradiation, evidence for NHEJ was portrayed as Takahashi et al. demonstrated that targeting the NHEJ repair pathway yielded higher radiosensitivity in cells than targeting HR, and Zhou et al. showed that inhibition of a central component in the NHEJ repair complex, DNA-PKcs, increased the radiosensitivity in cancer cells.[3,4] Evidence for HR was portrayed by Gerelchuluun et al., as they demonstrated HR was more involved in the repair of carbon ion-induced DSBs, and Wang et al., who demonstrated that high-LET radiation induces a large amount of short DSB fragments (<40 base pairs), which prevents the efficient loading and binding of the Ku heterodimer, a NHEJ factor, onto the broken DNA ends and leads to the suppression of NHEJ-mediated repair of the DSBs at these clustered damage sites.[5,6]

Carbon ion radiotherapy (CIRT) works because when carbon ion beams penetrate matter, they immediately start transferring their kinetic energy, quantified as linear energy transfer (LET). This increases as the carbon ion slows down until it depletes all its kinetic energy at the end of the carbon ion beam range, known as the Bragg peak, followed by the dramatic falling of the deposited dosage.[7] As the LET increases, the density of ionizations along the particle's traversal through the cell also increases, resulting in an increase in DNA damage complexity due to the formation of multiple close-proximity DNA damages including DSBs, single-strand breaks, and base damages. In clinics, this narrow Bragg peak can be extended to cover the entire tumor volume in the spread-out Bragg peak (SOBP) technique.[8] Therefore, the unique dose distribution of SOBP can achieve focused doses to tumor volume and minimize the dose to surrounding normal tissues, as the SOBP region is known to have higher doses at its proximal end and higher LET values at its distal end.

In the present study, we examined how the position within the 290-MeV/n carbon ion SOBP beam may influence the carbon ion-induced cytotoxicity and DSB distribution as well as the DSB repair pathway choice of NHEJ or HR using our previously established unique method of irradiation with our modified version of the cell survival assay, and utilizing γ-H2AX as a surrogate marker for DSB distribution.[9,10] Irradiation of the Chinese hamster ovary (CHO) wild-type cell line, 10B2, provided key insights into the cytotoxicity and DNA damage distribution while the established CHO-mutant cell lines, 51D1 and V3, defective for Rad51D and DNA-PKcs, in the HR-and NHEJ-DSB repair pathways, respectively, allowed us to decipher the importance of each of these DSB repair pathways at key depths within this range. This is because Rad51 paralog D (Rad51D) is an essential component in the HR pathway, as it is a mediator protein involved in the replacement of the replication protein A (RPA) with Rad51 to bind to the ssDNA overhang created in the initial steps of HR, which allows for the HR pathway to continue.[11] Furthermore, DNA-PKcs are known to play a key role in the NHEJ repair pathway, as they facilitate the rejoining reaction and their phosphorylation status indicates the level of kinase activity as well as the DNA end-binding ability.[12,13]

2 | MATERIALS AND METHODS

2.1 | Cell culture and irradiation conditions

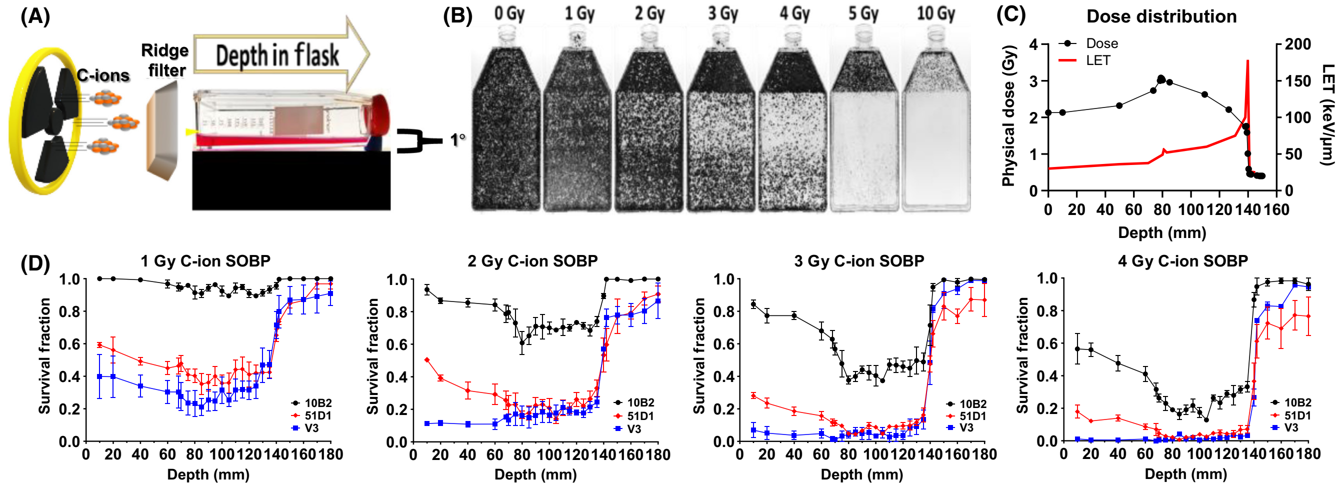
CHO wild type (10B2), DNA repair-deficient CHO mutants, V3 (DNA-PKcs),[14] and 51D1 (Rad51D)[15] cells were grown and maintained in α-MEM (Invitrogen) supplemented with 10% heat-inactivated fetal bovine serum (Sigma), antibiotics, and antimycotics (Anti-Anti; Invitrogen) at 37°C in incubators at 5% CO_2 and 100% humidity. Doubling times for CHO cell lines were approximately between 12 and 15 h. Carbon ions and Fe ions were accelerated to 290 and 500 MeV/nucleon, respectively, using the Heavy Ion Medical Accelerator in Chiba (HIMAC) synchrotron at the National Institutes for Quantum Science and Technology (QST formally NIRS), Chiba, Japan. Dose rates for carbon ions and Fe ions were set at 1 Gy/min. The irradiation field was within 2.5% uniformity.[16] Monoenergetic 290-MeV/n carbon ions and 500-MeV/n Fe ions had LET values of 13 and 200 keV/μm on entrance, respectively, and carbon ion beams were spread out with a ridge filter for 6 cm width of SOBP.[17] X-ray irradiation was performed at 200 kVp and 20 mA with aluminum (0.5 mm)–copper (0.5 mm) filters (Shimadzu, TITAN-320, NIRS), and dose rates were set at 0.5 Gy/min. Irradiations were carried out at room temperature. Beam characteristics and dosimetry using HIMAC have been described previously.[18,19] In brief, dosimetry of carbon ions was obtained by ionization chamber. LET values were sourced and referenced from prior studies by a calculation code, where fragmentation of nuclei was accounted for.[20,21]

2.2 | Irradiation procedure

Cells were irradiated in the same manner as described before.[9] Cell culture flasks or chamber slides were set up in a horizontal position (approximately 1°) to the carbon ion or Fe ion beam source, respectively, prior to irradiation.[22] The flasks rested flat on cell culture area, and the beam entry point was at the bottom of the flask (noncapped end; Figure 1A). Immediately following irradiation, CHO cells were incubated for a period of 8 days for colony formation, and directly following irradiation, flasks or chamber slides were incubated at 37°C and 5% CO2 for either 0.5 or 24 h prior to immunofluorescence staining.

2.3 | Survival fraction of cell survival assay calculation

Survival fractions were calculated as described previously.[9,10] Briefly, to quantify survival fractions at each evaluated depth for each cell line, they were scored for every millimeter along the width



Evaluated depth (mm)	10	80	85	115	125	140	150
Relative dose (Gy)	2.13	3.02	2.96	2.49	2.24	0.9	0.4

FIGURE 1 Cell survival versus depth of the full 290-MeV/n carbon ion spread-out Bragg peak (SOBP) beam range at increasing initial doses. (A) Illustration of the horizontal irradiation setup. (B) Images of CHO-10B2-WT T-175 cell culture flasks following increasing initial dose of carbon ion SOBP beam irradiation. (C) Dose and linear energy transfer (LET) distribution of carbon ion SOBP 290 MeV/n. Dose was measured via ionization chamber. LET values were referenced from prior studies.[20,21] (D) Comparison of survival fraction versus depth following increased initial carbon ion SOBP beam dosage between cell lines. Error bars indicated standard errors of the means from a minimum of three independent experiments per each initial dosage.

of the flask either possessing a surviving colony, defined as a colony containing >50 cells,[24] or not, and the average value was calculated. Therefore, survival fraction was calculated as number of colonies at specific depth divided by total millimeter along the width of the flask. Survival fractions for each depth, dosage, and cell line were then normalized with their respective control, reference at the 0-Gy dose, to account for differences in plating efficiencies. This approach was repeated for a minimum of three independent experiments per each initial treatment dosage of 0, 1, 2, 3, 4, 5, or 10 Gy per cell line.

2.4 | Sensitizer enhancement ratio (SER) calculation

D_{10} values (radiation dose to achieve 10% cell survival) were obtained from regression curves in either proximal (8.0–9.0 cm), mid (10.0–11.0 cm), or distal (12.0–13.0 cm) regions of SOBP. SER was calculated using D_{10} values of each cell line at each region of SOBP using the following formula:

$$\mathrm{SER}_{\mathrm{SOBP\ region}} = \frac{\mathrm{Average}D_{10}\mathrm{WT}_{\mathrm{SOBP\ region}}}{\mathrm{Average}D_{10}\mathrm{Deficient\ cells}_{\mathrm{SOBP\ region}}}$$

2.5 | Immunofluorescence staining

DNA double-strand break marker γ-H2AX foci formation assay was carried out as described previously.[9,23] Briefly, each slide containing cells was taken out of flask at either 0.5 or 24 h directly following the irradiation/incubation period, washed in cold PBS, fixed for 15 min in 4% w/v paraformaldehyde in PBS, washed in PBS, permeabilized for 5 min in 0.2% v/v Triton X-100 (Sigma) in PBS, washed twice in PBS, and treated with 10% goat serum for 1 h at 37°C for blocking. Antibodies were diluted with 10% v/v goat serum in PBS. Cells were incubated with 1:300 diluted mouse anti-γ-H2AX antibody (Millipore) for 1 h at 37°C, washed three times in PBS, incubated with 1:500 diluted Alexa Fluor488 goat anti-mouse IgG antibody (Abcam) for 1 h at 37°C, and washed four times in PBS. DAPI (4′,6-diamidino-2-phenylindole; Roche) diluted in SlowFade (Invitrogen, Thermo Scientific) was then applied to stain the DNA.

2.6 | γ-H2AX foci captured by Olympus FV3000 and analysis with BZ-X800 Keyence

Microscopic images were captured using an Olympus FV3000 confocal microscope (Olympus) and a ×100 objective lens (numerical aperture of 1.40). About 20–30 slices of images with every 0.33 μm were obtained to cover the CHO thickness to quantify γ-H2AX foci within the three-dimensional nucleus in a two-dimensional image. About 20–30 images were stacked into a single-layer image using cellSense imaging software (Olympus) for analysis of the γ-H2AX foci with BZ-X800 Analyzer software (Keyence). Each stacked image underwent haze reduction and automatic foci counting with BZ-X800 Analyzer software (Figure 2B). Foci scoring was carried out blindly in each experiment with at least 40 cells/each parameter analyzed. All foci scores were normalized with their respective scores in unirradiated cells (negative control). Unless stated otherwise, all foci analyses

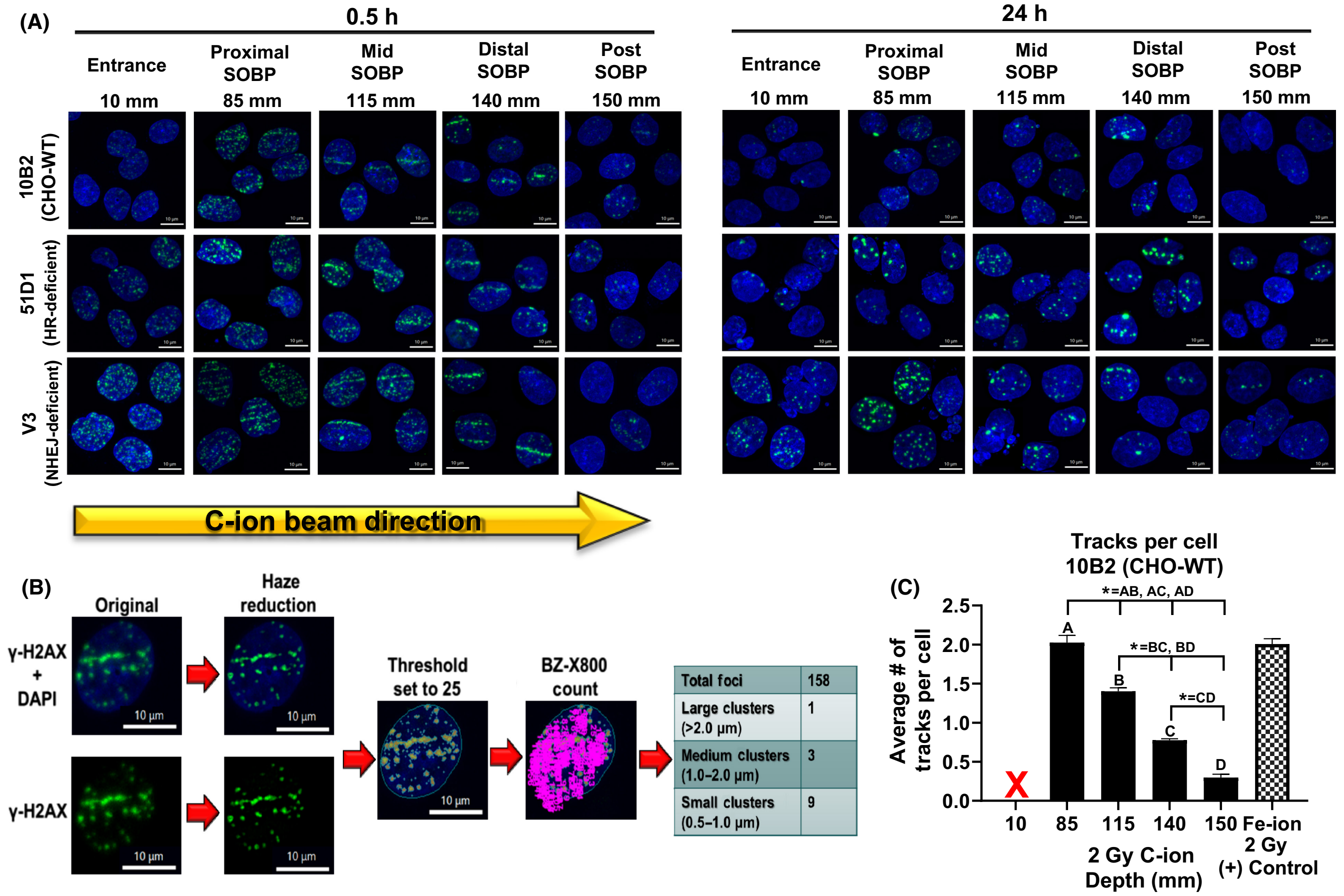


FIGURE 2 γ-H2AX foci formation and repair following an initial dose of 2 Gy 290 MeV/n carbon ion spread-out Bragg peak (SOBP) irradiation. (A) Representative images of carbon ion-induced foci at 0.5 h (left) and 24 h (right) following irradiation in each cell line. Analyzed depths for carbon ion SOBP beam were chosen from the prior clonogenic assays representing the entrance (10 mm), proximal SOBP (85 mm), mid SOBP (115 mm), distal SOBP (140 mm), and post SOBP (150 mm). Green indicates γ-H2AX foci. Blue is DAPI-stained nuclei. Indicator bar represents 10 μm. (B) γ-H2AX foci were automatically counted using BZ-X800 Analyzer software, and the results were obtained as total foci and by foci cluster sizes of small (0.5–1.0 μm), medium (1.0–2.0 μm), or large (>2.0 μm), as detailed in Materials and methods. (C) The average number of tracks per cell in CHO-10B2-WT cell lines at each analyzed depth following carbon ion SOBP and Fe ion beam irradiation. Multiple *t*-tests using the Holm–Sidak method; $*p < 0.05$. Error bars indicate standard errors of the means from a minimum of three independent experiments per initial dosage.

were reported followed by the mean and standard error of the means from at least three independent experiments. Total number of foci clusters with respective sizes within each individual cell was automatically reported. As previously reported, foci in close proximity to each other are likely counted as a single foci cluster (as was the case with the dense high-LET track structures).[24] Therefore, foci size within each cell was scored as small (0.5–1.0 μm), medium (1.0–2.0 μm), or large (>2.0 μm). Clear-tracked foci were scored by eye as a track in the CHO-10B2 wild-type cell lines. Intensity values reported by software were found to be unreliable for our experimental parameters and thus were not included.

2.7 | Statistical analysis

All experimental data were derived from at least three independent experiments. Statistical significance was determined by using multiple *t*-tests using the Holm–Sidak method and one-way analysis of variance (ANOVA) followed by the Bonferroni or Tukey multiple comparison test by GraphPad Prism 8 software (GraphPad). $p < 0.05$ was regarded as statistically significant difference for all tests.

3 | RESULTS

3.1 | Physical nature of 290 MeV/n carbon ion SOBP beam

All experimental flasks were irradiated in horizontal orientation to a 290 MeV/n carbon ion beam source whose Bragg peak was spread out to 6 cm width via a ridge filter (Figure 1A,B). When initial 2 Gy of SOBP carbon ions are irradiated in water, the dose reaches its peak value at 78.98 mm (3.08 Gy) and declines until it reaches the distal edge of the SOBP at 139.86 mm (1.01 Gy; Figure 1C). The post-SOBP tail region has a dose of approximately 0.4 Gy. Prior studies have reported dose-averaged LET values of approximately 40, 50, 75, and peak at 175 keV/μm at proximal-edge SOBP (84 mm), prox-mid SOBP (111 mm), distal-mid SOBP (131 mm), and distal-edge SOBP (140 mm), respectively.[20,21] In this study, increase in LET was biologically demonstrated by increases in foci cluster size of track structures within cells, as explained further below.

3.2 | Cytotoxicity within full range of 290 MeV/n SOBP carbon ion beam

All survival fractions were calculated and normalized with their respective controls as mentioned in Section 2 (Materials and methods). Consistent with our previous work, we observed a decrease in survival at our first evaluated depth (10 mm) with an increased initial beam entry irradiation dosage for all cell lines (Figure 1D; Figure S1A).[10] Under initial beam irradiation doses of 1, 2, 3, 4, 5, and 10 Gy at the entrance of the flask, the depths between 80 and 140 mm presented clear cytotoxicity with decreased survival fractions in the 10B2 (CHO-WT) cell line. This agrees with our prior work which defined the 290-MeV/n monoenergetic carbon ion Bragg peak at 140 mm and our ridge filter set for a 6-cm width SOBP.[9] Thus, 80–140 mm was defined as our biological SOBP range. Survival was not consistent throughout the SOBP region, as all CHO cell line derivatives (10B2, V3, and 51D1) demonstrated an increase in survival from the proximal to the distal SOBP region. These results demonstrate that the cytotoxicity is not equal across the SOBP possibly due to the separation of dose and LET in the carbon ion SOBP beam delivery technique.

3.3 | Cell survival of HR- and NHEJ-deficient cell lines

To further investigate the influence each DSB repair pathway has on cellular survival at each depth, the difference in the survival fraction mean values from 10B2 (CHO-WT) were compared for the 51D1 (HR-deficient) and V3 (NHEJ-deficient) cell lines (Figure S1B). V3 had a significantly higher difference from 10B2 following 2 and 3 Gy for depths from 10 to 60 mm and 10 to 40 mm, respectively ($p < 0.05$). Next, the SER for each cell line was calculated for the proximal (80–90 mm), mid (100–115 mm), and distal (120–130 mm) regions of SOBP (Table 1). In all regions of SOBP, the V3 SER value was significantly higher than 51D1 ($p < 0.05$). While SER values fluctuated between each region, there was a significant decrease in the SER values from the proximal to mid regions for both the 51D1 and V3 cell lines ($p < 0.05$). To determine the specificity to induce the most damage in the target SOBP region, the ratio of survival fraction at the entrance depth (10 mm) was compared with the lowest survival fraction within each region of the SOBP (Figure S1C). In the proximal region (80–95 mm), 51D1 was significantly higher than V3 and WT-10B2 following 2, 3, and 4 Gy initial irradiation doses ($p < 0.05$). In the mid region (100–115 mm), 51D1 was significantly higher than V3 and WT-10B2 following 2 and 3 Gy but only significantly greater than V3 following 4 Gy ($p < 0.05$). In the distal region (120–140 mm), 51D1 was higher than V3 following 2 Gy and both WT-10B2 and V3 following 4 Gy initial irradiation dose ($p < 0.05$). Collectively, these results demonstrate that the type of DSB repair mechanism can differentially influence the survivability and is dependent on the depth within the carbon ion SOBP beam range, that is, the dosage and/or LET deposition.

3.4 | DNA double-strand break distribution in the full range of 290-MeV/n carbon ion SOBP beam

We further investigated the results obtained in the clonogenic assays on a molecular level utilizing γ-H2AX as a surrogate marker for DSB distribution, as done previously.[9] DSB repair kinetics for each cell line were determined by analyzing the γ-H2AX foci at 0.5 and 24 h following 2 Gy initial irradiation dose of carbon ion SOBP beam, X-ray beam for positive low-LET control, or Fe ion beam for positive high-LET control (Figures 2A; Figure S2). At 0.5 h following carbon ion irradiation, all CHO cell line derivatives had a peak value in total of foci at 85 mm (proximal SOBP depth), which decreased with an increase in beam depth (Figure 3C). For 10B2-WT, this value was significantly higher than all other depths ($p < 0.05$). For 51D1 (HR-deficient), the value at 85 mm was significant to only depths outside of the SOBP (10 and 150 mm) where LET is known to be at its lowest ($p < 0.05$). At 24 h, 51D1's total foci increased throughout the SOBP, whereas 10B2 decreased. In contrast to 51D1 and 10B2, V3 (NHEJ-deficient) had high numbers of total foci at 0.5 h at entrance (10 mm) and was also the only cell line whose total foci decreased from entrance throughout the rest of the beam range at 24 h. V3 also demonstrated a significantly higher average total foci than 10B2-WT or 51D1 in the low-LET control, 2 Gy X-ray, for both 0.5 and 24 h (Figure 3A).

While the total number of foci at 0.5 h decreased across the SOBP, the average number of large foci per cell (>2.0 μm) at 0.5 h increased in 10B2-WT and 51D1 cells reaching a significant peak at 140 mm; V3 reached a significant peak at 115 mm ($p < 0.05$; Figure 3D). Taken together with Figure 2C, these results suggest that as the tracks/cell decrease, the density of the foci distribution within the tracks increases, creating large foci clusters. However, the number of large foci clusters increased from 0.5 to 24 h within the same depths for 10B2-WT at 85 mm, for 51D1 at 115 mm (with a peak at 140 mm), and for V3 at 10, 85, and 150 mm. Furthermore, NHEJ-deficient V3 and HR-deficient 51D1 had their highest number of large foci clusters at 24 h following the low-LET control 2 Gy X-ray and high-LET control 2 Gy Fe ion, respectively (Figures 3A,B). These results suggest that while high-LET track structures are apparent throughout all regions within the carbon ion SOBP, the tracks increase in foci cluster size with an increase in LET and that the DSB distribution at each beam depth differed depending on the DSB repair mechanism available and by cell type.

TABLE 1 D_{10} values and SERs within the SOBP.

Cell line	Proximal D_{10} (Gy)	Mid D_{10} (Gy)	Distal D_{10} (Gy)	Proximal SER	Mid SER	Distal SER
10B2 (CHO-WT)	4.67 ± 0.05	4.65 ± 0.16	5.10 ± 0.13	1.0	1.0	1.0
51D1 (HR-mutant)	2.87 ± 0.10	3.00 ± 0.19	3.25 ± 0.11	1.63	1.55	1.57
V3 (NHEJ-mutant)	2.71 ± 0.06	2.81 ± 0.05	3.06 ± 0.16	1.72	1.65	1.67

Note: D_{10} value = Average treatment dosage required to produce 10% survival in either the proximal (80–90 mm), mid (100–110 mm), or distal (120–130 mm) regions of the SOBP.

Abbreviations: CHO, Chinese hamster ovary; HR, homologous recombination; NHEJ, nonhomologous end joining; SER, sensitizer enhancement ratio; SOBP, spread-out Bragg peak.

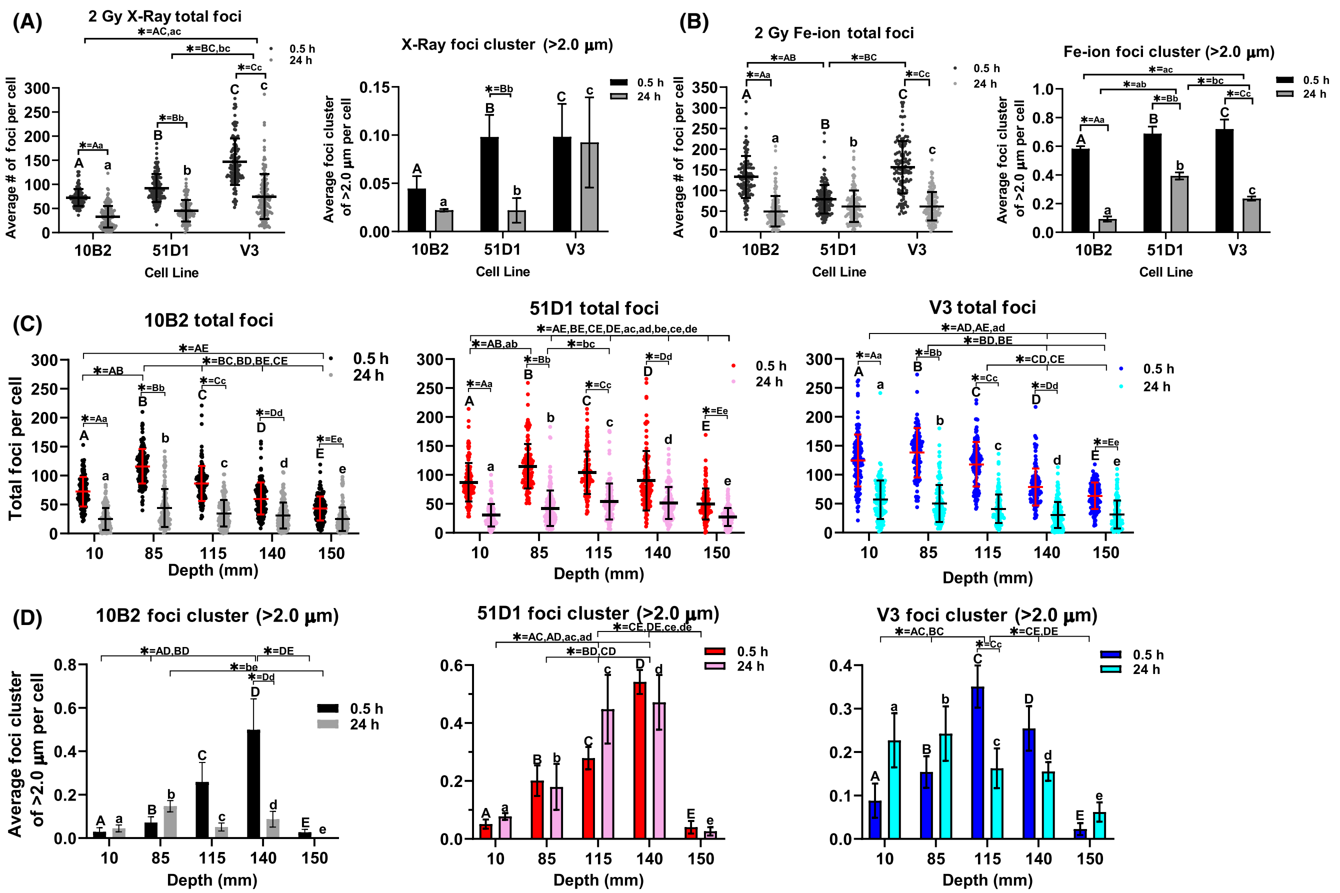


FIGURE 3 Average number of total foci and cluster size per cell are dependent on the depth within the carbon ion spread-out Bragg peak (SOBP) beam at 0.5 and 24 h following irradiation. (A) (left) Average number of total foci or (right) average number of large foci clusters (area > 2.0 μm) per cell in each cell line at 0.5 and 24 h following 2 Gy X-ray. (B) (left) Average number of total foci or (right) average number of large foci (>2.0 μm) per cell in each cell line at 0.5 and 24 h following 2 Gy Fe ion. (C) Average number of total foci per cell in each cell line following 2 Gy carbon ion SOBP. (D) Average number of large foci clusters per cell in each cell line following 2 Gy carbon ion SOBP. Statistical significance was determined between time points within the same depth by multiple *t*-tests using the Holm–Sidak method and between each depth by one-way ANOVA followed by Tukey's multiple comparison test. *$p < 0.05$. Error bars indicated standard errors of the means from a minimum of three independent experiments per initial dosage.

3.5 | DNA double-strand break repairability correlation with cellular survival

To address how the γ-H2AX foci distribution correlated with cellular survival, we compared survival to the repairability of the foci at each depth. Repairability was defined as the rate of change in the average number of total foci between 0.5 and 24 h (Figure 4A). For all cell lines this correlation was very high ($R^2 > 0.9$) except for 51D1 ($R^2 > 0.7$) because of the value at 115 mm. When repairability was compared between the cell lines, 51D1 (HR-deficient) was significantly lower at 115 and 140 mm (higher-LET regions) as compared with V3 (NHEJ-deficient; $p < 0.05$; Figure 4B). Following

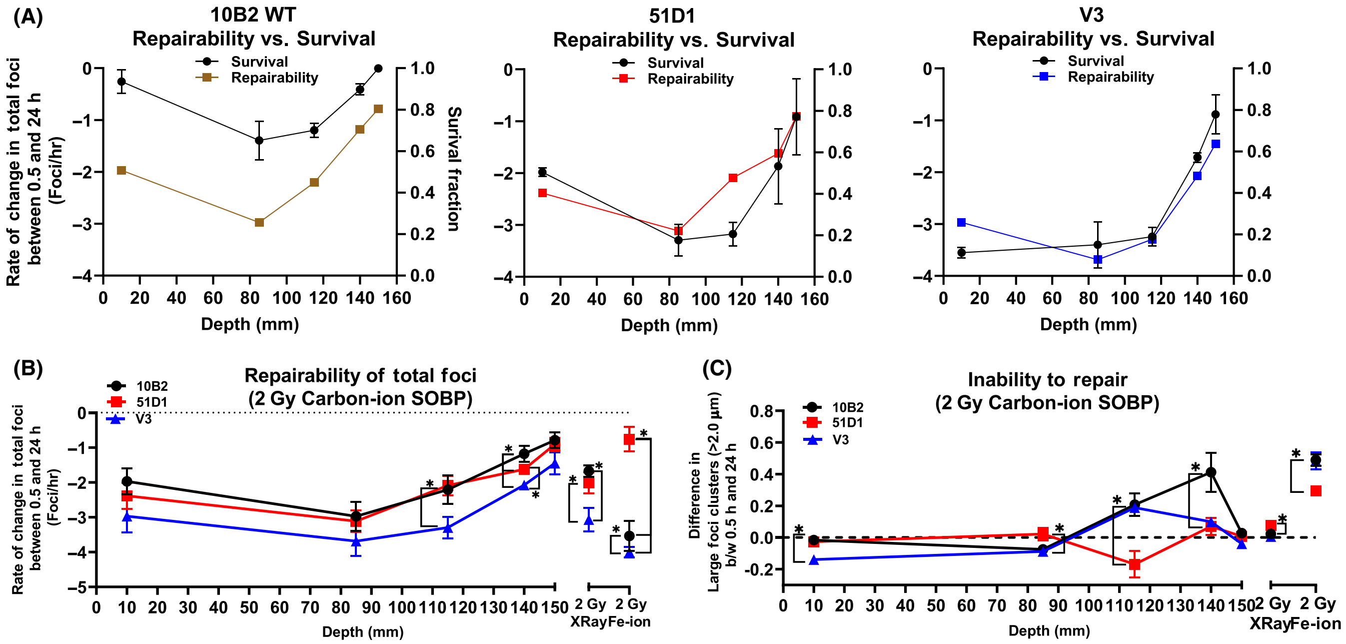


FIGURE 4 Repairability versus cell survival. (A) Correlation between rate of change in total foci from 0.5 to 24 h (foci/h) and survival fraction at each evaluated depth for each cell line. Simple linear regression; 10B2, V3 = $R^2 > 0.9$; 51D1 = $R^2 > 0.7$. (B) Comparison of repairability of total foci in each cell line at each evaluated depth and positive controls. (C) Comparison of each cell line's change in average number of large foci clusters (foci >2.0 µm) from 0.5 to 24 h at each evaluated depth and positive controls. Negative values indicate an increase in the number of large foci clusters at 24 h. One-way ANOVA followed by Tukey's multiple comparisons test; *$p < 0.05$. Error bars indicated standard errors of the means from a minimum of three independent experiments per initial dosage.

low-LET control, 2 Gy X-ray, V3 (NHEJ-deficient) was significantly higher than both 10B2-WT and 51D1 (HR-deficient; $p < 0.05$), which may be due to the alt-NHEJ DSB repair mechanism being capable of repairing these damages. Following high-LET control, 2 Gy Fe ion, 51D1 (HR-deficient) repairability was significantly lower than that of all other cell lines ($p < 0.05$).

Next, we compared each cell line's inability to repair (ITR). ITR) was defined as the change in large foci clusters (>2.0 µm) between 0.5 and 24 h, as prior studies have demonstrated that unrepaired DSBs would result in larger foci when viewed at later time points (Figure 4C).[24] V3 (NHEJ-deficient) was least able to repair from entrance to proximal SOBP (10–85 mm). 51D1 (HR-deficient) was significantly less able to repair at mid and distal SOBP (115 and 140 mm) and following 2 Gy Fe ion ($p < 0.05$; Figure S3). These results demonstrate that V3 (NHEJ-deficient) was less capable of repairing the damage from entrance to the proximal SOBP (higher-dosage, lower-LET regions), while 51D1 (HR-deficient) was less capable of repairing the damage from mid to distal SOBP (lower-dosage, higher-LET regions), which highly correlated with the cellular survival results.

4 | DISCUSSION

In this study, we investigated the DSB repair pathway choice for repairing carbon ion-induced DSBs at increasing SOBP beam depths, as this has been a long-standing area of debate. To understand the importance behind each pathway choice within the carbon ion SOBP beam range, we first had to investigate how the beam distributes its biological damage. While prior studies have suggested a uniform biological dose throughout the carbon ion SOBP, defined as the dose from combination of the physical dose with the RBE, our cell survival results demonstrated nonuniform cytotoxicity.[25,26] From proximal to distal SOBP, all CHO cell lines demonstrated an increase in cell survival. γ-H2AX analysis of the CHO-10B2-WT cells demonstrated that DSB distribution was dependent on beam depth, as the highest number of foci/cell was observed at proximal SOBP (highest dosage region), but the highest number of large foci clusters was observed at distal SOBP (highest LET region). In a prior study, foci cluster size within carbon ion-induced track structures of hTERT cells was observed to increase with an increase in LET.[27] Our results agree, as 0.5 h following carbon ion SOBP irradiation we observed an increase in foci cluster size most notable within the high-LET track structures (i.e., number of large foci clusters/cell) from proximal to distal SOBP, therefore, biologically representing an increase in LET.

However, we could not definitively correlate the large foci clusters with tracks at 24 h due to the possibility of chromatin reorganization in our cells, which may have led to incorrect assignment of residual track structures per cell, even though prior studies demonstrated the presence of these track structures following varying sources of high-LET radiation at 24 h in 48BR, 2BN, and hTERT cell lines.[24,28] Instead, we found that large foci clusters were also a quality method to determine the cells' inability to repair DSB at 24 h following irradiation, as a prior study demonstrated unrepaired damage will result in larger foci at this time point.[24] We observed the highest number of large foci clusters at proximal SOBP in CHO-10B2-WT cells, which correlated well with the lowest survival being at this depth. Collectively, these results also portray the overkill effect in distal SOBP, in which a single particle deposits much more energy than required to kill a cell and results in killing less cells per absorbed dose. Prior studies have also observed at very high LET values that the percentage of non-hit cells increased, which helps explain why we observed an increase in survival at distal SOBP within our CHO cell lines.[28]

While many studies have attempted to determine which DSB repair pathway is most important for carbon ion-induced DSBs, each reports conflicting results possibly due to their experimental setup. Each study compared either the DSB pathway choice of differing incident ions which increased in their incident LET values, such as photons, protons, carbon-, oxygen, and Fe ions, or incident carbon ions of specific LET values.[3-6] While the methods utilized in these studies are of common practice, much information is lost, such as the effects from the actual dose and/or LET distributions in different beam depths. By directly comparing the dose and/or LET contribution to each DSB repair pathway utilizing a carbon ion SOBP beam in a single biological system, we were able to address how the importance of each pathway changed according to the beam depth and utilized positive controls of low LET (X-ray) and high LET (Fe ion) to support our findings.

Our results demonstrated that both DSB repair-defective CHO-mutant cell lines, 51D1 (HR-deficient) and V3 (NHEJ-deficient), had increased radiosensitivity as compared with CHO-10B2-WT cells. However, it was apparent that NHEJ was most essential in the higher-dosage, lower-LET depths from entrance leading up to the SOBP. Importantly, V3 demonstrated a dose-dependent shift in radiosensitivity, as survival fraction fell to nearly zero following 2 Gy initial dose at entrance but increased throughout the SOBP. These results suggested this shift was dose but not LET dependent, and cell death was likely the result of large numbers of "simple" DSBs, as the complexity of DSBs is known to increase with LET.[29-31] This reasoning is consistent with a prior study, which demonstrated that less complex DSBs are primarily repaired through NHEJ and the efficiency of NHEJ in repairing DSBs diminishes as the DSB complexity increases and the preferred DSB repair pathway becomes HR.[32] γ-H2AX foci analysis further supported our survival results, as V3 cells had the highest average number of total foci/cell at 0.5 h and were significantly less capable of repairing the damage at entrance and proximal SOBP. V3 also had the highest number of total foci following both of our positive controls but was significantly less capable of repairing the damage following low-LET positive control, 2 Gy X-ray.

On the other hand, our results suggest HR is most essential within higher LET depths of the SOBP. HR-deficient cell lines, 51D1, had the highest entrance to lowest survival fraction ratios in the proximal, mid, and distal regions of the SOBP, which is a key property required for a radiosensitizer. In contrast to V3, at 24 h 51D1 also demonstrated an increase in total foci/cell and large foci clusters throughout SOBP. Moreover, 51D1 was significantly less capable of repairing the damage in the high-LET mid and distal regions of SOBP as well as in high-LET control, 2 Gy Fe ion, meaning HR is essential for the repair of high LET-induced DSBs.

Collectively, these results demonstrate a potential benefit for inhibiting HR via RAD51 over the inhibition of NHEJ because inhibition of NHEJ would severely enhance the DNA damage to the surrounding healthy tissue outside of the SOBP. This strategy has also been very appealing, as prior reports demonstrated that HR inhibition via Gleevec-mediated inhibition of Rad51 expression and Rad51 siRNA may promote preferential sensitization of tumor cells relative to normal cells.[33-35] Several potential drugs that have had promising results are RAD51-inhibitory compounds, such as RI-1, RI-2, BO2, IBR2, and another that is currently anonymous due to patent application disclosing structures.[36-39] In future studies, it would be beneficial to utilize our system to address the effectiveness of drugs, such as RI-1, in combination with CIRT in human cell lines. Specifically, in cancers such as pancreatic cancer that has shown promising results with high-LET CIRT in combination with Gemcitabine, which promotes apoptosis of cells undergoing DNA synthesis.[40-42]

The main drawback of our system of irradiation comes with the large quantities of cells required for each γ-H2AX experiment (9×10^6 cells per T175 cell culture flask), currently limiting our choice of cell types which can be utilized, and the DNA damage response can be strongly dependent on the cell type. While the CHO cell lines utilized in this study have been well established and these experiments were built on our prior studies,[9,10,43,44] it is important to note that CHO cells are known to have genomic instability. Recently, Spahn et al. investigated the relationship between DSB repair and genome instability in CHO cells and demonstrated that this genome instability can be improved by correction of DNA repair genes, particularly in NHEJ.[45] This may have contributed to our observed

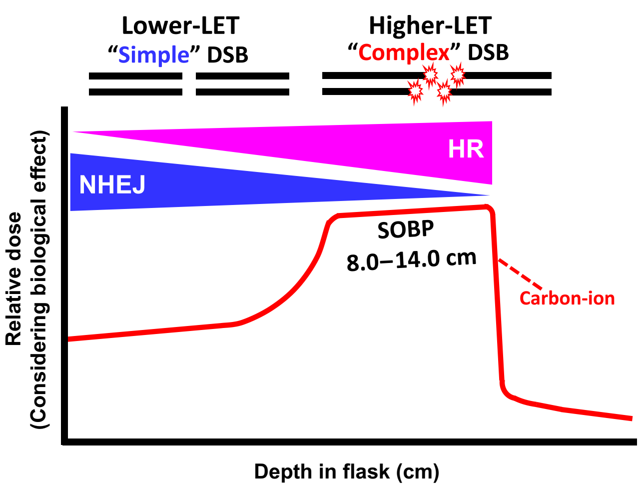


FIGURE 5 Graphical representation demonstrating the importance of each DNA double-strand breaks (DSB) repair pathway throughout the carbon ion spread-out Bragg peak (SOBP) beam range. Nonhomologous end joining (NHEJ) is more important in depths prior to the SOBP. Homologous recombination (HR) is more important in the depths within the SOBP. LET, linear energy transfer.

lowest survival fractions being at the proximal region of the SOBP in our CHO-WT (10B2) cell line, which further supports that NHEJ is important in that region of the SOBP, but HR becomes more important, as the LET increases throughout the SOBP. However, it is necessary to expand our results into human cell lines by further optimizing our experimental parameters to accommodate the unrealistically high cost of materials for these experiments, such as the amount of siRNA required for DSB repair pathway inhibition that would be needed to treat the large quantities of cells used in each experiment. Nevertheless, the results of this study also lay the foundation for these future studies in human cell lines, as they have identified key depths of interest within the full carbon ion SOBP beam range, which will allow us to further optimize our system to focus only on cells within these key depths.

In conclusion, NHEJ is important throughout the full carbon ion SOBP range, but the importance of NHEJ decreases while the importance of HR increases as the beam proceeds throughout the SOBP (Figure 5). Simply put, NHEJ-mediated repair correlated more closely with dosage, while HR-mediated repair correlated more closely with LET. Therefore, HR inhibition is a potentially optimal target to complement CIRT, and these results may contribute to the development of more effective treatment methods.

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CONFLICT OF INTEREST STATEMENT

None.

ETHICS STATEMENT

Approval of the research protocol by an Institutional Review Board: N/A.

Informed Consent: N/A.

Registry and the Registration No. of the study/trial: N/A.

Animal Studies: N/A.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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